

# nRF54Lxx 蓝牙 BIS 时间同步

微秒级多节点时间同步 · 技术方案与演示

BLE 5.2 ISO

1 发 2 收

<1 $\mu$ s 精度

200Hz 更新

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Nordic nRF Connect SDK v3.4.0 · nRF54L15

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# 演示目标

## OBJECTIVE

### 多节点统一时间基准

1 个发送端 + 2 个接收端，长期共享同一时间基准

### 微秒级打戳一致

三方对同一事件打出相同时间戳，精度约  $<1\mu\text{s}$

### 硬件级确定性呈现

P1.10 输出  $1\mu\text{s}$  窄脉冲，边沿对齐  
可被逻辑分析仪直观观测

# 什么是 BLE BIS

## BROADCAST ISOCHRONOUS STREAM · 规范解读

### 定义与规范依据

- BIS = Broadcast Isochronous Stream  
广播等时流，BLE 5.2 引入的  
LE Isochronous Channels ( LE-ISO ) 特性
- 多条 BIS 组成一个 BIG  
( Broadcast Isochronous Group )，  
一对多广播，无需连接/回传
- 规范：Bluetooth Core Spec  
Vol 6 Part G ( Isochronous  
Adaptation Layer, ISOAL )
- 核心机制：以固定 ISO\_Interval 发送 SDU，  
BIG 锚点作为同步参考时间基准

### 关键参数与同步流程

关键参数：ISO\_Interval ( 5ms~4s，  
步进 1.25ms ) · SDU\_Interval · RTN · NSE  
Max Transport Latency · 2M PHY

1

Broadcaster 发 periodic adv  
( 携带 BIGInfo 控制信息 )

2

Sync Receiver 扫描并  
同步 Periodic Advertising

3

按 BIGInfo 建立 BIG 同步

4

定时接收 SDU，BIG 锚点  
作为时间基准

# 技术原理

PRINCIPLE · 一页看懂

BIS 广播 (5ms 间隔)

发送端 TX  
SDU = [trigger][counter][tx\_ts]

接收端 RX  
解析 → 时钟伺服 → shared\_ts

## ① 时间戳载体

发送端把 SDU 同步参考时间戳  
tx\_ts 写入 payload; 接收端拿到  
同一瞬间的本地读数 info->ts

参考对 (tx\_ts, info->ts)

## ② 时钟伺服

$\text{offset} = \text{发送端时钟} - \text{接收端时钟}$   
中位数估计 + ppm 漂移估计  
外推到当前时刻

$\text{shared} = \text{local} + \text{offset} + \text{drift} \times \text{age}$

## ③ 定时呈现

GRTC → DPPI → GPIOTE 硬件链路  
在 presentation 时刻驱动 P1.10  
免软件线程抖动

GRTC → DPPI → GPIOTE

关键: tx\_ts 与 info->ts 是同一物理瞬间 (BIG 锚点) 在两个自由运行时钟下的读数 → 据此估计偏移与漂移

# 系统架构

ARCHITECTURE · 1 发 2 收



# Demo 环境

SOFTWARE · HARDWARE · TOOLS

## 硬件 Hardware

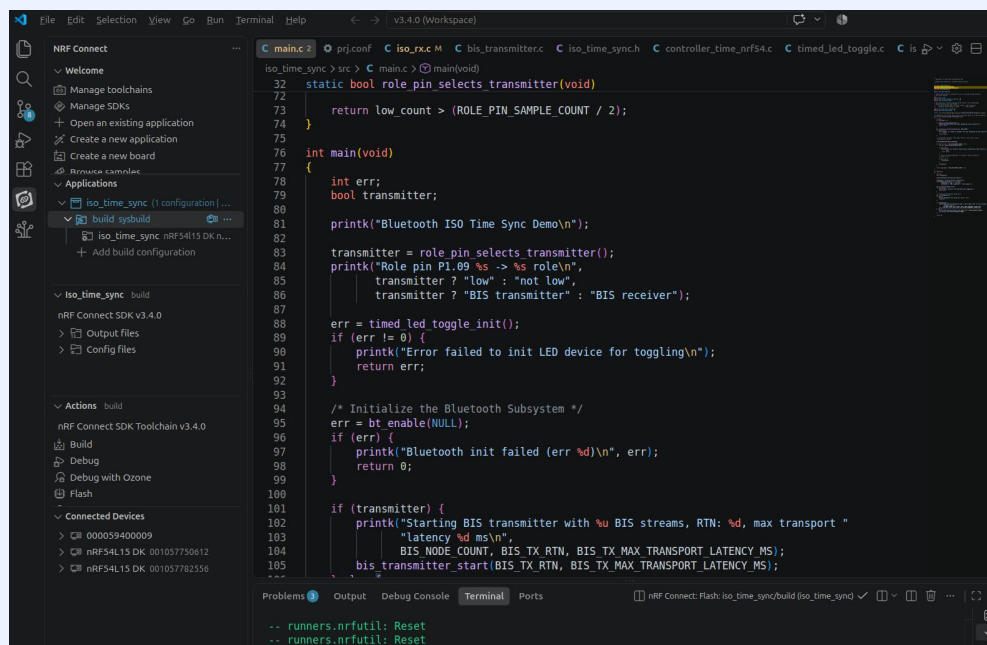
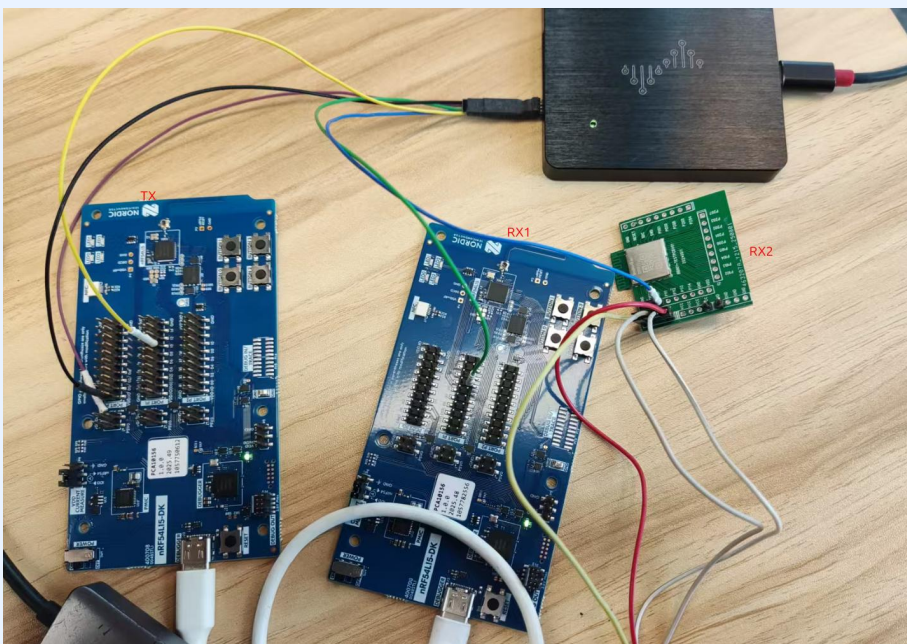
- nRF54L15-DK × 3 ( 1 发 + 2 收 )
- USB 数据线 × 3
- 逻辑分析仪 ( ≥ 20MS/s )
- 杜邦线 / 测试探针

## 软件 Software

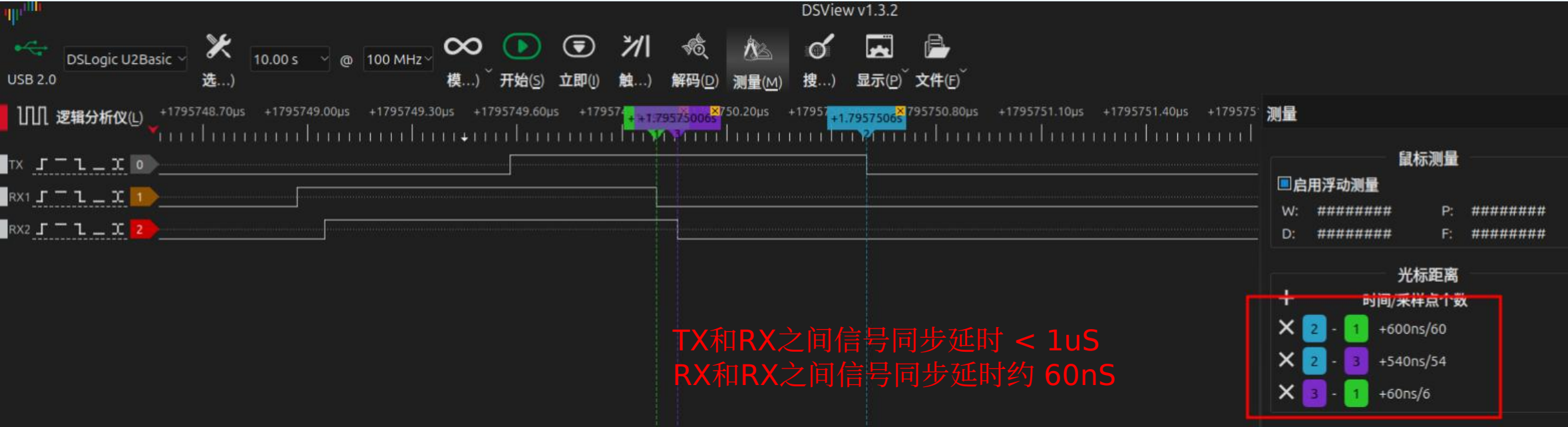
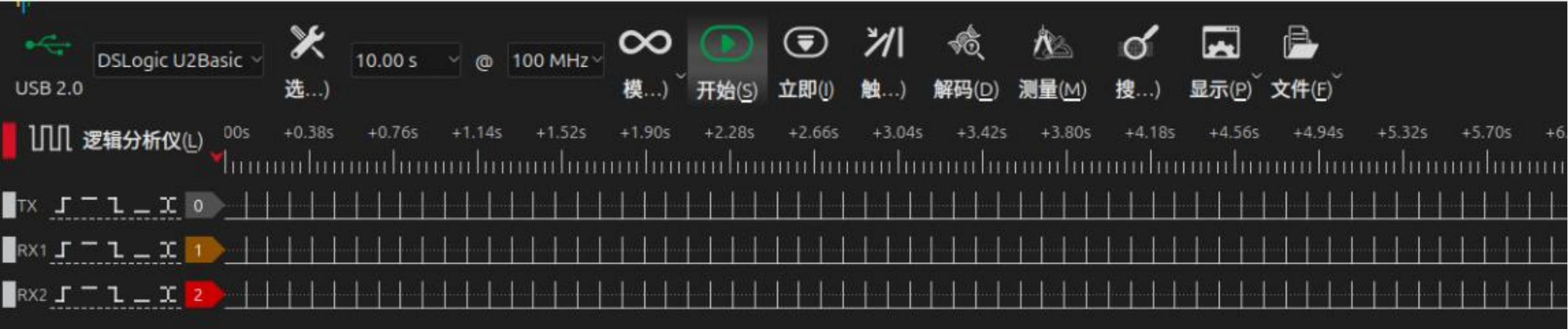
- nRF Connect SDK v3.4.0
- Zephyr RTOS 4.4
- 本工程固件 zephyr.hex
- 编译: west build

## 工具 Tools

- nrfutil ( 烧录 / 复位 )
- 串口终端 ( PuTTY / minicom )
- 逻辑分析仪软件 ( Saleae 等 )



# 实测数据1---逻辑分析仪测量IO口同步脉冲



# 实测数据2--- log观察TX和RX在相同counter时的时间戳

```
TX
devin@devin-HP-Pavilion-Laptop-14-dv0xxx: ~
: 1 LED will be set in 2365 us
Sent SDU counter 1428900 with timestamp 2849696111 us, controller_time 2849695554 us, btn_val
: 1 LED will be set in 2367 us
Sent SDU counter 1429000 with timestamp 2850196111 us, controller_time 2850195553 us, btn_val
: 1 LED will be set in 2368 us
Sent SDU counter 1429100 with timestamp 2850696111 us, controller_time 2850695559 us, btn_val
: 1 LED will be set in 2362 us
Sent SDU counter 1429200 with timestamp 2851196111 us, controller_time 2851195555 us, btn_val
: 1 LED will be set in 2366 us
Sent SDU counter 1429300 with timestamp 2851696111 us, controller_time 2851695567 us, btn_val
: 1 LED will be set in 2354 us
Sent SDU counter 1429400 with timestamp 2852196111 us, controller_time 2852195576 us, btn_val
: 1 LED will be set in 2345 us
Sent SDU counter 1429500 with timestamp 2852696111 us, controller_time 2852695559 us, btn_val
: 1 LED will be set in 2362 us
Sent SDU counter 1429600 with timestamp 2853196111 us, controller_time 2853195557 us, btn_val
: 1 LED will be set in 2364 us
Sent SDU counter 1429700 with timestamp 2853696111 us, controller_time 2853695554 us, btn_val
: 1 LED will be set in 2367 us
Sent SDU counter 1429800 with timestamp 2854196111 us, controller_time 2854195553 us, btn_val
: 1 LED will be set in 2368 us
Sent SDU counter 1429900 with timestamp 2854696111 us, controller_time 2854695560 us, btn_val
: 1 LED will be set in 2361 us
Sent SDU counter 1430000 with timestamp 2855196111 us, controller_time 2855195556 us, btn_val
: 1 LED will be set in 2365 us
Sent SDU counter 1430100 with timestamp 2855696111 us, controller_time 2855695566 us, btn_val
: 1 LED will be set in 2355 us
Sent SDU counter 1430200 with timestamp 2856196111 us, controller_time 2856195565 us, btn_val
: 1 LED will be set in 2356 us
Sent SDU counter 1430300 with timestamp 2856696111 us, controller_time 2856695562 us, btn_val
: 1 LED will be set in 2359 us
Sent SDU counter 1430400 with timestamp 2857196111 us, controller_time 2857195546 us, btn_val
: 1 LED will be set in 2375 us
Sent SDU counter 1430500 with timestamp 2857696111 us, controller_time 2857695550 us, btn_val
: 1 LED will be set in 2371 us
Sent SDU counter 1430600 with timestamp 2858196111 us, controller_time 2858195574 us, btn_val
: 1 LED will be set in 2347 us

RX1
devin@devin-HP-Pavilion-Laptop-14-dv0xxx: ~
us
Recv SDU counter 1429000 timestamp 1635109032 us btn_val: 1 shared_ts 2850196111
us
Recv SDU counter 1429100 timestamp 1635609045 us btn_val: 1 shared_ts 2850696111
us
Recv SDU counter 1429200 timestamp 1636109057 us btn_val: 1 shared_ts 2851196111
us
Recv SDU counter 1429300 timestamp 1636609070 us btn_val: 1 shared_ts 2851696111
us
Recv SDU counter 1429400 timestamp 1637109082 us btn_val: 1 shared_ts 2852196111
us
Recv SDU counter 1429500 timestamp 1637609095 us btn_val: 1 shared_ts 2852696112
us
Recv SDU counter 1429600 timestamp 1638109107 us btn_val: 1 shared_ts 2853196111
us
Recv SDU counter 1429700 timestamp 1638609119 us btn_val: 1 shared_ts 2853696111
us
Recv SDU counter 1429800 timestamp 1639109132 us btn_val: 1 shared_ts 2854196111
us
Recv SDU counter 1429900 timestamp 1639609144 us btn_val: 1 shared_ts 2854696111
us
Recv SDU counter 1430000 timestamp 1640109157 us btn_val: 1 shared_ts 2855196111
us
time_sync: locked=1 offset=1215086970 us drift=-25 ppm samples=256 max_resid=17
us
Recv SDU counter 1430100 timestamp 1640609169 us btn_val: 1 shared_ts 2855696111
us
Recv SDU counter 1430200 timestamp 1641109181 us btn_val: 1 shared_ts 2856196111
us
Recv SDU counter 1430300 timestamp 1641609194 us btn_val: 1 shared_ts 2856696111
us
Recv SDU counter 1430400 timestamp 1642109206 us btn_val: 1 shared_ts 2857196111
us
Recv SDU counter 1430500 timestamp 1642609219 us btn_val: 1 shared_ts 2857696111
us
Recv SDU counter 1430600 timestamp 1643109231 us btn_val: 1 shared_ts 2858196111
us
```

TX: Sent SDU counter  
1430600  
timestamp  
2858196111 us

RX: Recv SDU counter  
1430600  
shared\_ts  
2858196111 us

→ 完全一致

时间戳一致误差

<1 μs

# 技术优势

## ADVANTAGES

### 微秒级精度

- controller-timed 硬件呈现 ( GRTC/DPPI/GPIOTE )
- 时钟伺服 + 漂移外推, 消除陈旧误差
- 实测时间戳一致误差  $<1\mu\text{s}$

### 长期稳定可靠

- 在线估计晶振漂移 ( ppm ), 持续跟踪
- 失同步 holdover 按最后速率滑行
- 32 位时间戳回绕正确处理

### 轻量易扩展

- 一对多广播, 无连接、无回传
- 接收端数量可扩展
- O(1) 伺服、无动态内存、200Hz 开销可忽略

### 硬件级确定性

- GPIO 呈现免软件线程调度抖动
- SET/CLR 绝对电平、幂等自愈
- 中位数估计, 对丢包/重传鲁棒

# 应用场景

## SCENARIOS

### 分布式采集

多传感器/多设备  
精密时间戳对齐

### 工业控制

状态监测与  
协同控制

### 声学/振动

多通道同步  
采集分析

### 协同测量

统一时间基准的  
测量系统

# 总结

基于 BLE BIS + 控制器时间戳 + 时钟伺服，实现微秒级多节点时间同步

实测时间戳完全一致 ( $<1\mu\text{s}$ ) · 脉冲边沿亚 $\mu\text{s}$  级(600nS)对齐 · 长期稳定 · 一对多可扩展

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感谢观看 · 欢迎交流